ANDRÉ ROGÉRIO DA SILVA FILHO

Software Consultant & Applied AI Engineer | Full-Stack · Autonomous Agents · Cloud Architecture

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PROFESSIONAL SUMMARY

Software Consultant and Applied AI Engineer with a background spanning enterprise systems, corporate IT infrastructure, and production software architecture. Specializes in building deterministic, resilient systems around autonomous AI agents, vision-language models, and modern full-stack platforms using TypeScript, Python, React, Next.js, Node.js, and PostgreSQL. At CBI, acts as software consultant and solutions architect leading internal platform engineering, high-availability database systems (PostgreSQL/Cloud SQL) safeguarding sensitive citizen records across metropolitan public infrastructure concessions, and autonomous AI agents (Google Gemini, MCP servers, advanced RAG). Combines rigorous cloud governance with hands-on software delivery to engineer high-scale, production-ready systems.

TECHNICAL SKILLS

Languages & Core: TypeScript, Python, JavaScript, SQL, HTML5, CSS3/TailwindCSS.

Frontend Engineering: React 19/18, Next.js 15, Three.js, React Three Fiber, WebGL/GLSL Shaders, Responsive UI/UX Architecture.

Backend & Microservices: Node.js, Express, Fastify, RESTful APIs, WebSockets, Microservices, Zod Schema Validation.

Applied AI & Agents: Multi-Agent Systems, LangGraph, Model Context Protocol (MCP), LLMs (Google Gemini API), Vision-Language Models (Qwen2-VL via vLLM on AMD ROCm), PyTorch, Advanced RAG (Hybrid Search & Reranking), NetworkX Relational Graphs, Structured JSON Outputs.

Cloud & Databases: Google Cloud Platform (Cloud Run, Cloud SQL, Secret Manager), PostgreSQL (Row-Level Security), Supabase, Redis, Prisma ORM, Docker, CI/CD, Git.

Security & Enterprise Governance: LGPD/GDPR Compliance, AES-256-GCM Encryption, IDOR Protection, Microsoft 365, Entra ID, Microsoft Graph API, Microsoft Intune Telemetry, Certified AppSec Practitioner (CAP).

PROFESSIONAL EXPERIENCE

CBI – Companhia Brasileira de Infraestrutura
São Paulo, SP, Brazil (Remote) • 06/2026 – Present

Software Consultant & Solutions Architect (10/2026 – Present)

- Lead enterprise software architecture and internal platform engineering, developing scalable full-stack web applications and AI-driven automation pipelines using TypeScript, React 19, Python, and PostgreSQL.
- Architect and manage high-availability cloud infrastructure on Google Cloud Platform (Cloud Run, Cloud SQL, Secret Manager) and relational databases, directly responsible for safeguarding sensitive citizen data across metropolitan public concessions in Brazil.
- Direct technical engagements and architecture alignment with enterprise platform vendors (including Google Cloud and Microsoft), ensuring rigorous LGPD compliance, zero-trust access, and infrastructure resilience.
- Conceived, architected, and deployed **"Isa"**, an enterprise AI assistant utilizing Google Gemini, autonomous tool execution via Model Context Protocol (MCP) servers, contextual RAG retrieval, and deterministic JSON Schema validation with Zod.
- Lead corporate AI technical enablement, conducting hands-on engineering workshops on secure LLM deployment patterns and prompt-to-code architectures.

IT Systems Analyst (06/2026 – 10/2026)

- Managed enterprise identity governance, access security, and endpoint fleet telemetry across Microsoft 365, Entra ID, and Microsoft Intune.
- Engineered automated security and compliance monitoring pipelines using Microsoft Graph API and Intune, advancing overall infrastructure posture to 89% and IAM posture to 91%.
- Developed custom automation scripts in Python and PowerShell to optimize cross-departmental operations and incident deflection, resulting in promotion to Software Consultant.

Farmamed Drogarias
Sorocaba, SP, Brazil • 10/2025 – 03/2026

IT Systems & Automation Developer

- Developed Python automation microservices and scripts for multi-branch log analytics, systems synchronization, and automated incident diagnosis.
- Administered enterprise identity provisioning, access control, and endpoint governance utilizing Active Directory, Entra ID, and Microsoft Intune.
- Monitored core operational services and resolved high-priority technical incidents within strict business SLAs.

Wobben Windpower
Sorocaba, SP, Brazil • 11/2023 – 04/2025

Software & IT Systems Intern

- Selected for the global **Microsoft Copilot Flight Crew** program, evaluating generative AI enterprise features and real-world adoption workflows.
- Developed internal automation scripts and integration routines for SAP ERP, Active Directory, and Microsoft 365 in a heavy industrial manufacturing environment.
- Provided tier-2 technical resolution for corporate service desk incidents via ServiceNow under strict SLA benchmarks.

SELECTED PROJECTS

Cribum (AuditRx) – Multimodal Medical Audit & Fraud Graph github.com/TheyDreez/cribum

`AMD Developer Hackathon (Health Track) • Python, FastAPI, React 19, PyTorch, AMD ROCm, vLLM, Qwen2-VL, NetworkX, TypeScript`

- Architected a local, privacy-first multi-agent medical claim and prescription audit system built for the AMD Developer Hackathon.
- Integrated local Open Vision-Language Models (Qwen2-VL served via vLLM on AMD ROCm) with uncertainty-aware multi-hypothesis extraction and clinical LASA safety databases.
- Engineered a relational multigraph anomaly detection engine using NetworkX, identifying systemic medical billing collusion and synthetic prescription fraud rings.

Onboarding OS – Enterprise Multi-Tenant Lifecycle Platform github.com/TheyDreez/onboarding-os-portfolio

`React 19, TypeScript, Node.js, Express, PostgreSQL, Supabase, Redis, Zod, AES-256-GCM, DocuSeal`

- Engineered a multi-tenant SaaS platform automating employee onboarding workflows connecting HR, IT, and managers across municipal subsidiaries (Grupo CBI).
- Implemented strict tenant isolation via PostgreSQL Row-Level Security (RLS) and transparent AES-256-GCM encryption at rest for sensitive documents.
- Built automated daily background workers for LGPD data lifecycle compliance (anonymization upon retention expiry) and state-machine-validated document pipelines.

Memory Lens – Cognitive Computing Platform & 3D Neural Interface github.com/TheyDreez

`TypeScript, React 19, Three.js, React Three Fiber, WebGL/GLSL, Express, Gemini API, Vitest`

- Designed and developed a cognitive computing platform modeling human memory retention, associative recall, and structured forgetting (Ebbinghaus curve), reducing LLM context overhead by up to 80%.
- Engineered a 3D bioluminescent neural organism ("Chief Organism") in React Three Fiber with custom GLSL shaders synchronized to cognitive state telemetry via an event bus.
- Conducted as formal academic research on semantic document organization under Prof. Dr. Fabrício Londero (UniFacens), targeting submission to the ACDSA international conference.

FundiVR – Industrial Digital Twin & VR Training Simulator github.com/TheyDreez/fundivr-training-simulator

`React 18, TypeScript, Fastify, PostgreSQL, Prisma ORM, WebSockets, Docker, Vitest`

- Architected an industrial digital twin training simulator for aluminum furnace operators in 700°C+ environments, featuring operational safety state machines.
- Engineered real-time bi-directional telemetry via WebSockets with an automated scoring engine evaluating SOP compliance and reaction time.
- Built a decoupled zero-dependency demo mode enabling standalone presentation without external database connectivity.

TECHNOLOGY RESIDENCY

ICT Resident Researcher (Digital Twins & 5G) 10/2026 – Present

UniFacens • Softex • Ministry of Science, Technology and Innovation (MCTI) | Residency (TIC 51)

- Selected among the **top 20 out of 300 participants** from the ICT Specialization (360h) to join the Technological Residency (6-month track), building industrial digital twins, real-time telemetry, and private 5G networks.
- Conceptualized and engineered the FundiVR digital twin prototype, defended before an academic and industrial evaluation panel.

EDUCATION & ACADEMIC BACKGROUND

Centro Universitário SENAC Expected Graduation: 2026

Higher Technology Degree in Systems Analysis and Development (TADS) • São Paulo, Brazil

CERTIFICATIONS & LANGUAGES

Certifications: Google Cloud Generative AI Leader • Certified AppSec Practitioner (CAP) • CCNA 1: Introduction to Networks (Cisco) • Network Security & Endpoint Security (Cisco).

Languages: English: Advanced / Fluent (C1+) • Portuguese: Native • Spanish: Intermediate (Technical Communication).

Volunteering: Volunteer English Teacher at Clube Poliglota Sorocaba, mentoring international students across cross-cultural settings.